A Brief History of Air Pollution and Exposure Pathways

ENVH 111

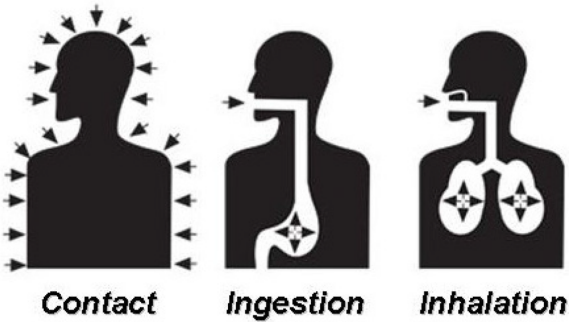
Eddie Kasner

Overview

1. Air pollution sources, exposure routes and pathways
2. Historic air quality events
   - Meuse Valley, Belgium (1930)
   - Donora, Pennsylvania (1948)
   - Great Smog of London (1952)
   - Los Angeles and Clean Air Act (1970)
   - Beijing “Airpocalypse” (2013-2014)
   - Recent wildfire smoke events (2020-2025)
3. Air pollution studies
4. Government and policy

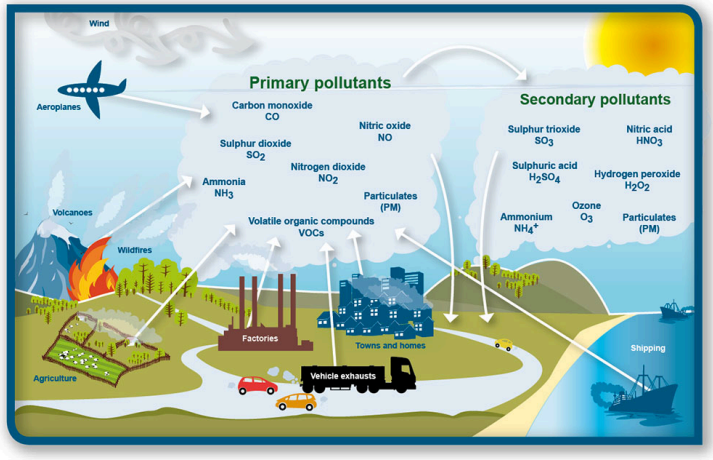
Exposure Pathways and Routes

- Exposure pathway = source + medium + receptor
- Exposure route = how a contaminant enters receptor
  - Dermal contact
  - Ingestion
  - Inhalation



Damalas C. 2016. Farmers' Exposure to Pesticides: Toxicity Types and Ways of Prevention. Open Access Toxics 2016, 4(1): 1.

Air Pollution Sources



Scotland's Environment. Air quality. https://www.environment.gov.scot/our-environment/air/air-quality/

Exposure Pathway

- Sources
  - Natural: volcanic eruptions, wildfires, allergens
  - Human: energy use, transportation, industry, agriculture
- Pollutants
  - Primary: hazardous substance emitted directly into atmosphere
    - CO - gas, coal, or wood combustion
    - NO_x - fossil fuel combustion, e.g. brown haze around cities
    - SO_2 - volcanoes and industrial coal combustion
    - PM - particulate matter, e.g. fossil fuel combustion, brake dust, smoke
    - NH_3 - agricultural processes, e.g. fertilizer
    - VOCs - volatile organic compounds, e.g. benzene, toluene
    - Pb - "anti-knocking" agent formerly added to gasoline
  - Secondary: hazardous substance formed in atmospheric reaction
    - O_3, SO_3, HNO_3, H_2SO_4, H_2O_2, NH_4^+, PM
- Receptors: humans and animals (each pollutant has different health effect)

Meuse Valley Fog, Belgium (1930)

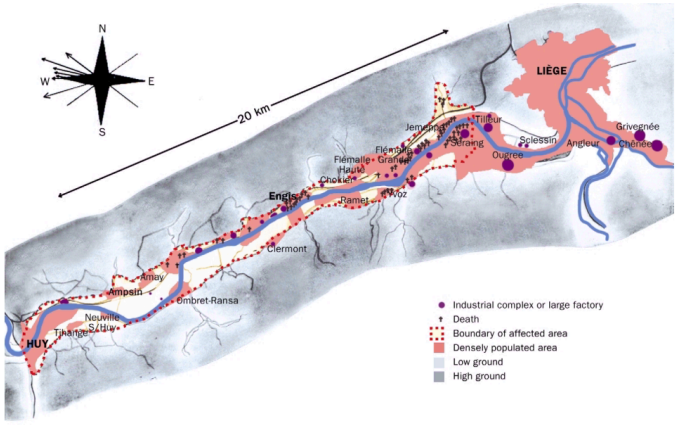


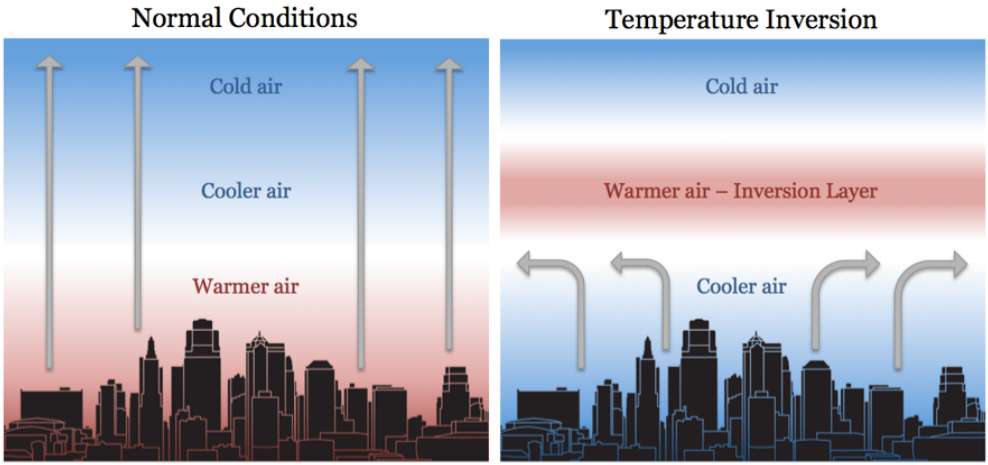
Figure 1: **Map of the Meuse Valley between Liège and Huy, indicating the fog-covered area and location of fatalities and factories**
Reproduced and modified from figure 1 of Firket and colleagues' report.[10]

Nemery et al. 2001. The Meuse Valley fog of 1930: an air pollution disaster. The Lancet 357(9257): 704-708.

Meuse Valley Fog, Belgium (1930)

- Sources
  - SO_2, H_2SO_4, or HF from nearby zinc, glass, and steel factories
  - Inversion: warm air layer trapped cold, polluted air in the valley (5 days in Dec)
- Receptors
  - Asthma-like symptoms: shortness of breath, coughing fits
  - Estimated 60 excess deaths (20-89 years old)
- Impacts
  - Seen at the time as "an unavoidable consequence of prosperity"
  - Interviewed family physicians and patients days after smog
  - First scientific proof of link between air pollution and illness and death
  - Demonstrated role of temperature inversions

Temperature Inversion



Understory captures evidence of temperature inversion in Kansas City. http://understoryweather.com/understory-captures-evidence-of-temperature-inversion-in-kansas-city/

Donora, Pennsylvania Smog (1948)



Donora Historical Society and Smog Museum. https://www.sites.google.com/site/donorahistoricalsociety/1948-smog

Donora, Pennsylvania Smog (1948)

- Sources
  - CO, SO_2, and H_2SO_4 from steel and zinc factories
  - Temperature inversion (5 days in Oct)
- Receptors
  - About 14,000 residents trapped under thick layer of smog
  - Nearly 7,000 experienced respiratory or cardiovascular issues
  - About 20 deaths during the event
- Impacts
  - Widely publicized around world and people acted
  - Scientists began to study link between air pollution and health
  - Helped lead to first US federal air law: Air Pollution Control Act (1955)

Great Smog of London (1952)



A tugboat on the Thames near Tower Bridge in heavy smog, 1952. Fox Photos, Getty Images. https://timedotcom.files.wordpress.com/2016/11/london-fog.jpg

Great Smog of London (1952)

- Sources
  - Cold weather led residents to burn extra coal [SO_2, PM] to stay warm
  - Inversion: 30-mile stagnant air mass (5 days in Dec)
- Receptors
  - About 100,000 residents felt ill with respiratory symptoms
  - Estimated 4,000-12,000 excess deaths, mostly from bronchitis and pneumonia
- Impacts
  - Especially bad "pea-souper" as seen on TV (e.g. *The Crown*)
  - Led to UK Clean Air Act of 1956 (and 1970 in US)
  - Landmark in air pollution epidemiology: first detailed analysis linking pollutant levels to illness and death (Nemery et al. 2001)

Los Angeles Smog (1954)



Highland Park Optimist Club wearing smog-gas masks at banquet, Los Angeles, c. 1954. UCLA Library.

Los Angeles Smog and Clean Air Act (1970)

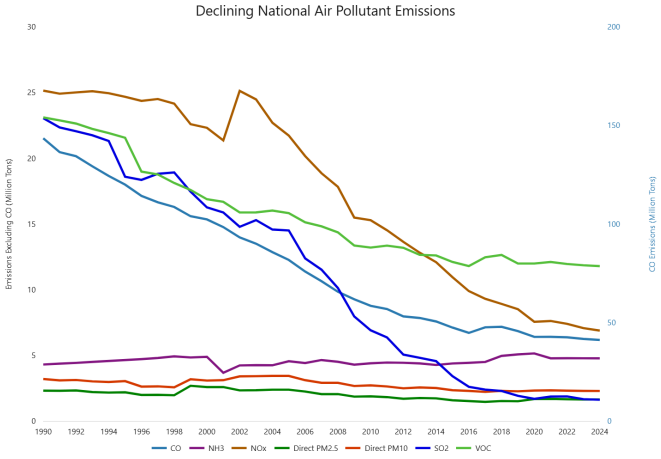


Photochemical smog in Los Angeles. Getty images.

Los Angeles Smog and Clean Air Act (1970)

- Sources
  - 1,000,000+ vehicles on LA roads by 1940
  - VOC emissions + sunlight + $NO_x \rightarrow O_3$ (photochemical smog)
  - LA is natural trap for smog accumulation
    - Geography: flat basin between ocean and mountains
    - Weather patterns: cold ocean current creates inversions
- Receptors
  - Respiratory and cardiovascular effects among residents
  - Systematic spatial association between respiratory mortality and increased O_3 levels in 1961 (Mahoney 1976)
- Impacts
  - Clean Air Act of 1970 set federal and state regulations to limit stationary and mobile source emissions
  - O_3 levels far below 1970s peaks despite many more cars
  - Still ranked worst US metro for O_3 in 26 of last 27 years (ALA State of the Air 2026)

US Emissions After the Clean Air Act (1990-2024)



US EPA. Our Nation's Air 2025. https://gispub.epa.gov/air/trendsreport/2025/

Beijing “Airpocalypse” (2013-2014)

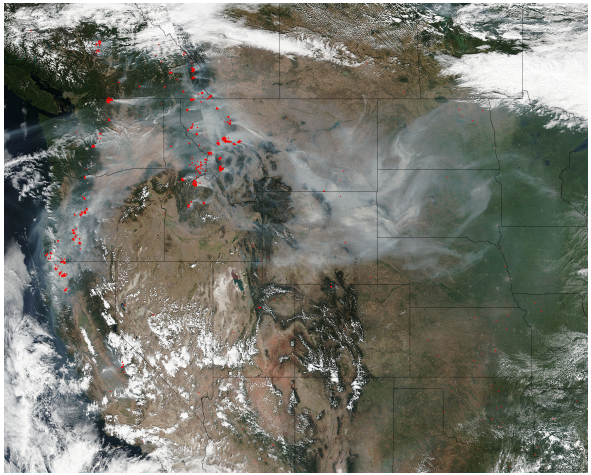


Beijing smog vs. no smog. 2014. Getty Images.

Beijing "Airpocalypse" (2013-2014)

- Sources
  - PM from burning coal for heat in winter, industry, and vehicle traffic
  - Like LA, Beijing in "natural bowl" with temperature inversions
- Receptors
  - Respiratory and cardiovascular effects among residents
  - Jan 2013: $PM_{2.5}$ went "beyond index," over 20x WHO guideline
- Impacts
  - 2013 Air Pollution Action Plan: cut coal, closed old plants, cleaner fuels
  - Annual $PM_{2.5}$ fell from about 100 $\mu g/m^3$ (2013) to 27 $\mu g/m^3$ (2025)
  - US Embassy hourly air quality posts raised public pressure

West Coast Wildfire Smoke



Smoke and wildfires in western U.S., 2017. NASA. https://www.nasa.gov/image-feature/goddard/2017/smoke-and-fires-light-up-pacific-northwest

Recent Wildfire Smoke Events (2020-2025)

- Sources
  - Hotter, drier conditions and strong winds drive larger fires
  - Smoke (mostly $PM_{2.5}$) travels hundreds to thousands of miles
- Receptors
  - Entire regions, rural and urban, often far from the fire
- Some recent major wildfire smoke events
  - West Coast, Sept 2020: Portland had world's worst major-city air; daily AQI peaked at 357 (Portland) and 196 (Seattle)
  - East Coast, June 2023: smoke from Canadian fires pushed NYC AQI near 500; about 62 million people exposed
  - Los Angeles, Jan 2025: urban fires destroyed 16,000+ structures; smoke and ash carried heavy metals
- Impacts
  - Protection relies on AQI alerts, indoor air filtration, and limiting time outdoors

Air Pollution Studies

1. Harvard "Six Cities" Study
2. Multi-Ethnic Study of Atherosclerosis and Air Pollution (MESA Air), led by UW
3. UW Adult Changes in Thought Air Pollution (ACT-AP) Study: $PM_{2.5}$ and dementia
4. State of Global Air 2024: air pollution linked to 8.1 million deaths (2021), 2nd leading risk factor
5. Burke et al. 2023, *Nature*: wildfire smoke erased about 1/4 of $PM_{2.5}$ gains since 2000 in 35 states
6. Local monitoring you can check today:
   - Puget Sound Clean Air Agency sensor map, WA Ecology air quality map, EPA AirNow Fire and Smoke Map
7. Health effects commonly studied: mortality, respiratory disease and lung cancer, cardiovascular disease and stroke, low birthweight, dementia

Government and Policy

1. Clean Air Act (1956-UK; 1970-USA)
2. National Ambient Air Quality Standards (NAAQS)
   - CO, Pb, NO_2, O_3, PM_{10}, $PM_{2.5}$, SO_2
   - 2024: annual $PM_{2.5}$ standard lowered from 12 to 9 $\mu g/m^3$ (upheld in federal court, 2026)
3. WHO Air Quality Guidelines (2021): annual $PM_{2.5}$ guideline of 5 $\mu g/m^3$
4. Air quality modeling and monitoring
   - Dispersion and receptor modeling; Puget Sound Clean Air Agency and WA Ecology monitors
   - EPA Our Nation's Air 2025: national air quality and emissions trends
5. WA wildfire smoke rule for workers (L&I, 2024; updated 2025)
6. WA Climate Commitment Act: improving air quality in 16 overburdened communities